DSAA 4011 · LAB 6 · FRIDAY 16 OCTOBER 2026

Teaching a Model to Reason

Way 2 of Lecture 6 inside a reasoning language model

A model is trained on questions and final answers only. Today: how it can still learn the **steps** between, in one tiny maze.

Yuwen Huang
Advanced Machine Learning and Deep Learning · HKUST (Guangzhou)

Builds on Lecture 6: When the E-step is too big

Outline

- **The problem: the steps are hidden**

The model: a walker in a maze

The method: walk, keep, count

What it learns, and where it fails

The same loop in today's models

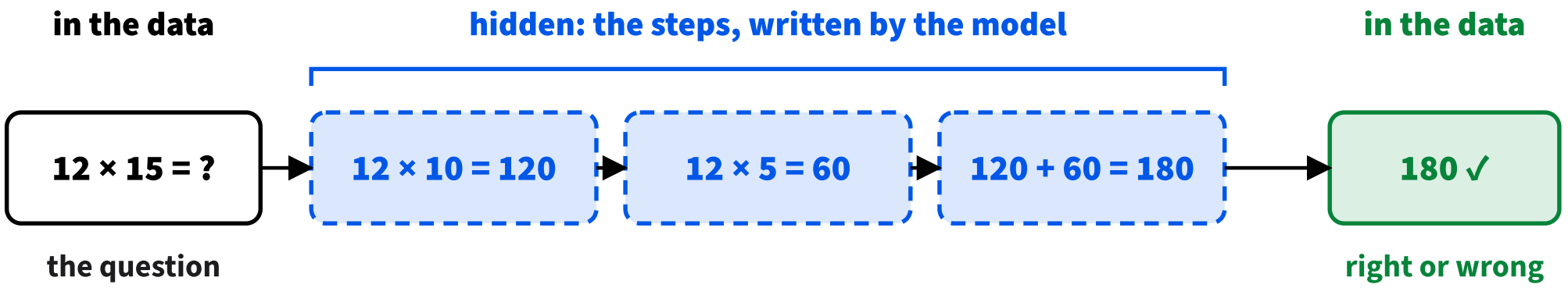
One question runs through the lab: how can a model learn steps it never sees?

Main Question

A reasoning model is trained on questions with answers only. Where does its reasoning come from?

1 · The problem: the steps are hidden

Trouble: **the data shows the question and the answer, not the steps.**

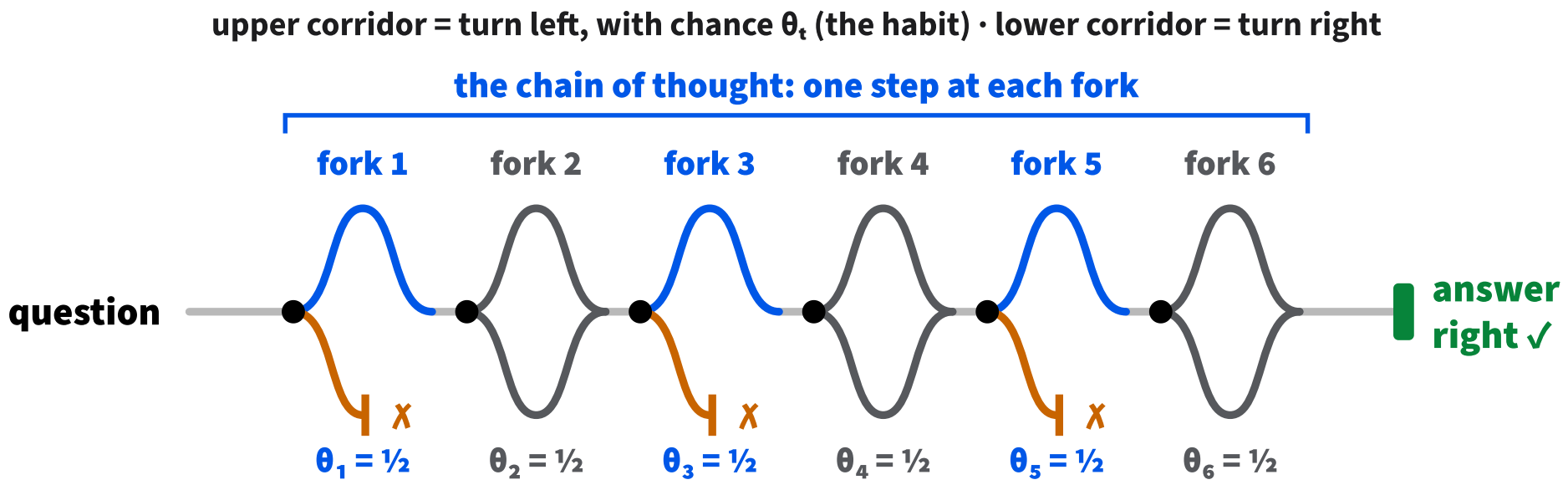


a real chain of thought has hundreds of steps, and each could have been written another way

Look: the steps play the part of Lecture 5's hidden coins, so to learn good steps we must first **guess** them, as expectation–maximization (EM) does.

2 · The model: a walker in a maze

Each fork is one reasoning step; the exit is a right answer.



Look: the six habits $\theta_1, \ldots, \theta_6$ are the model's **weights**. Our goal: habits that make the exit likely.

3 · The method: what EM needs

Trouble: the guess step weighs every path, and there are 2^L.

GUESS STEP for every path z, its chance given the exit, p(z | exit).

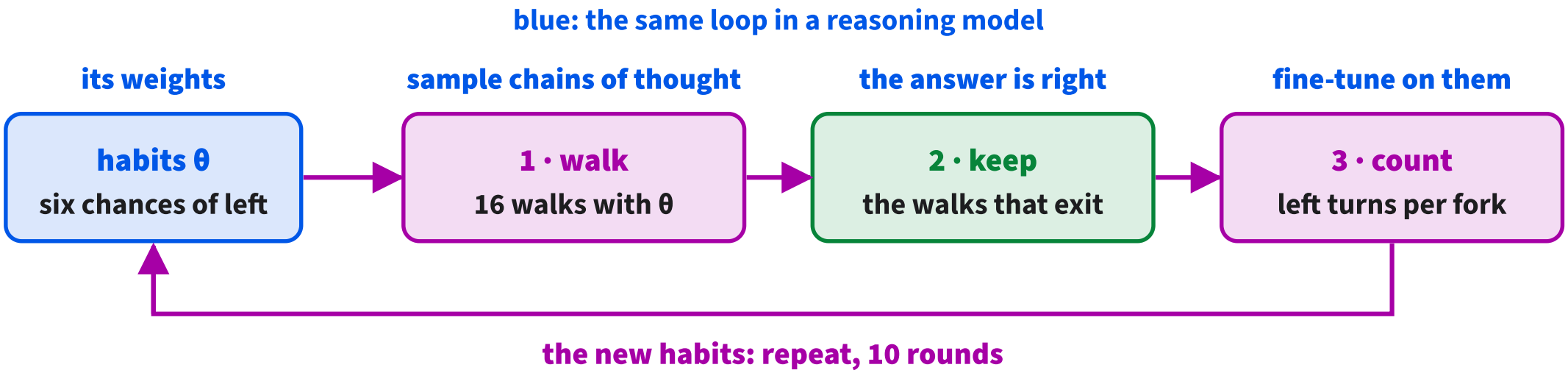
COUNT STEP count the turns at each fork, each path weighted by that chance.

steps L	paths 2^L	
6 (our maze)	64	a computer lists them at once
20	about a million	a short chain of thought
30	about a billion	a long chain of thought

Read: two choices at every step double the list each time, so for a real chain of thought the guess step cannot be written out. This is expectation–maximization (EM) [Dempster, Laird and Rubin, 1977].

3 · The method: walk, keep, count

Fix (Way 2): walk, keep the walks that exit, count the turns.



count, one imaginary walk each way: $\theta_t \leftarrow$ (left turns at fork t among kept + 1) / (kept + 2)

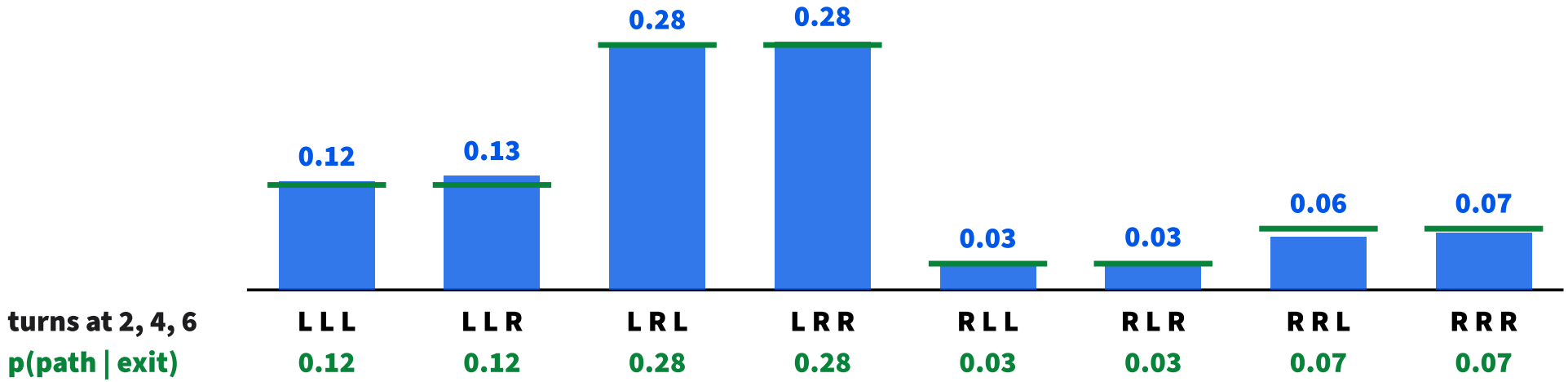
Read: no list of paths: we only run the walker forward and look at the exit. This is Monte Carlo expectation–maximization (EM) [Wei and Tanner, 1990].

3 · The method: why the count is right

The kept walks follow p(path | exit), the guess step's chances.

$$\Pr(\text{a kept walk is path } z) = \frac{p(z) \cdot \mathbf{1}[z \text{ exits}]}{\Pr(\text{exit})} = p(z \mid \text{exit})$$

habits: forks 1, 3, 5 at ½ · θ_2 = 0.8, θ_4 = 0.3, θ_6 = 0.5 · 16 000 walks, 2 028 kept



blue bar: share among the kept walks · green line: p(path | exit), computed from the habits

Read: a path is walked with chance p(z) and kept only if it exits, so among the kept walks it appears with chance p(z | exit). Counting kept walks is therefore the count step of expectation–maximization (EM), plus sampling noise **(Lecture 6, p. 29)**.

4 · Round 1 by hand

Sixteen walks with every habit at ½; then keep and count.

walk	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
fork 1	L	R	L	L	R	R	R	R	L	R	L	R	L	R	L	R
fork 2	R	L	R	R	R	R	L	L	R	L	L	L	L	R	L	L
fork 3	R	R	L	R	R	L	L	R	L	L	L	R	L	L	R	R
fork 4	R	R	L	R	L	L	L	L	L	R	L	L	L	R	L	L
fork 5	L	R	R	L	L	L	R	L	L	R	L	R	R	L	R	L
fork 6	L	R	L	R	R	L	L	R	L	L	L	R	L	L	R	R
exit?	✗	✗	✗	✗	✗	✗	✗	✗		✗		✗	✗	✗	✗	✗

kept: 2 of 16 walks exit (green) · fork 1: left in 2 of 2 → $\theta_1 = (2 + 1) / (2 + 2) = 0.75$

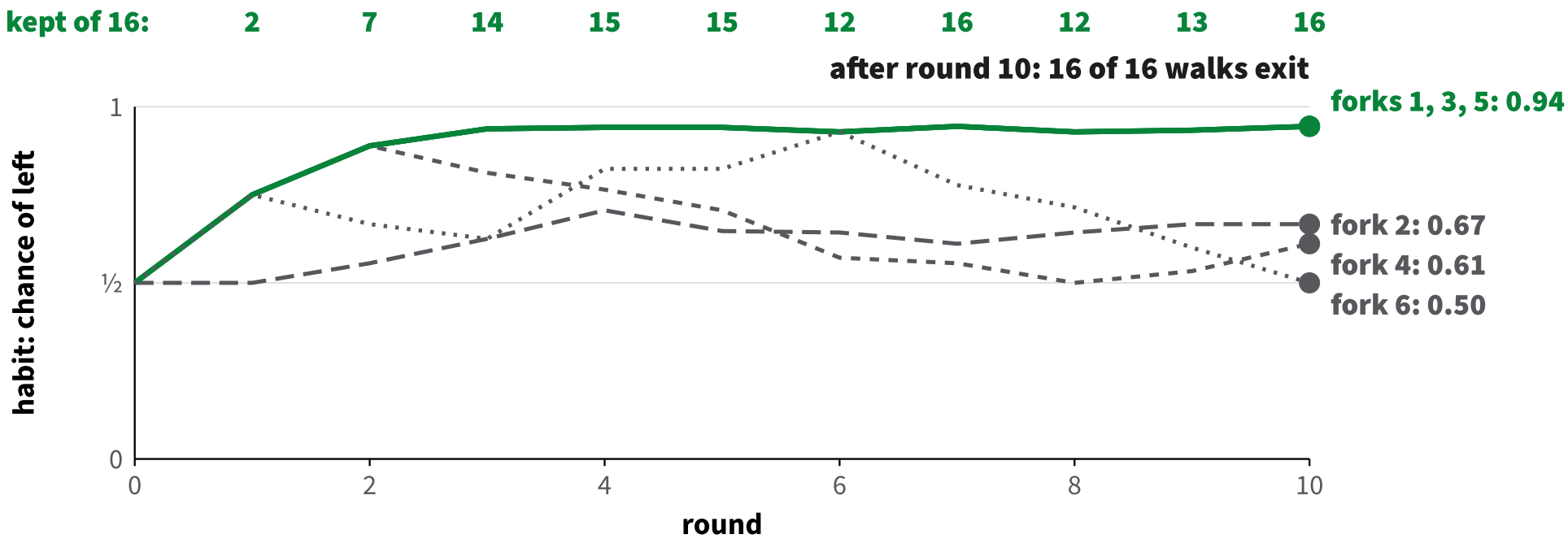
fork 2: left in 1 of 2 → $\theta_2 = (1 + 1) / (2 + 2) = 0.50$ · the same count at every fork

Watch: each column is one walk; an orange R at fork 1, 3 or 5 is a dead end; the green columns are kept. One imaginary walk each way keeps a habit off 0 and 1.

Q: How many of the 16 walks exit? A: About 2: a walk needs left at three forks, ½ · ½ · ½ = 1 in 8, and 16 / 8 = 2.

4 · Ten rounds

Result: the forks that matter learn; the free forks only wander.



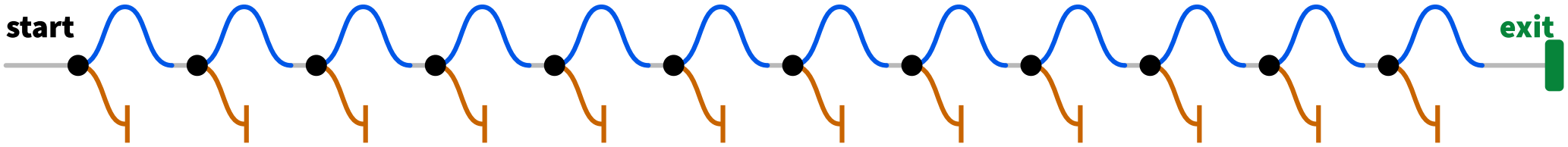
Watch: the kept walks rise from 2 to 16 of 16. Exact expectation–maximization (EM) over all 64 paths ends at 0.93 and ½ **(p. 21)**: sampling agrees, plus noise.

Q: Why do the free forks not learn? A: The exit never tells a left turn there from a right one: the model learns only what the answer check can see.

4 · Where it fails: a cold start

Trouble: twelve forks, no walk exits, nothing to count.

12 forks that all matter: the exit needs a left turn at every one



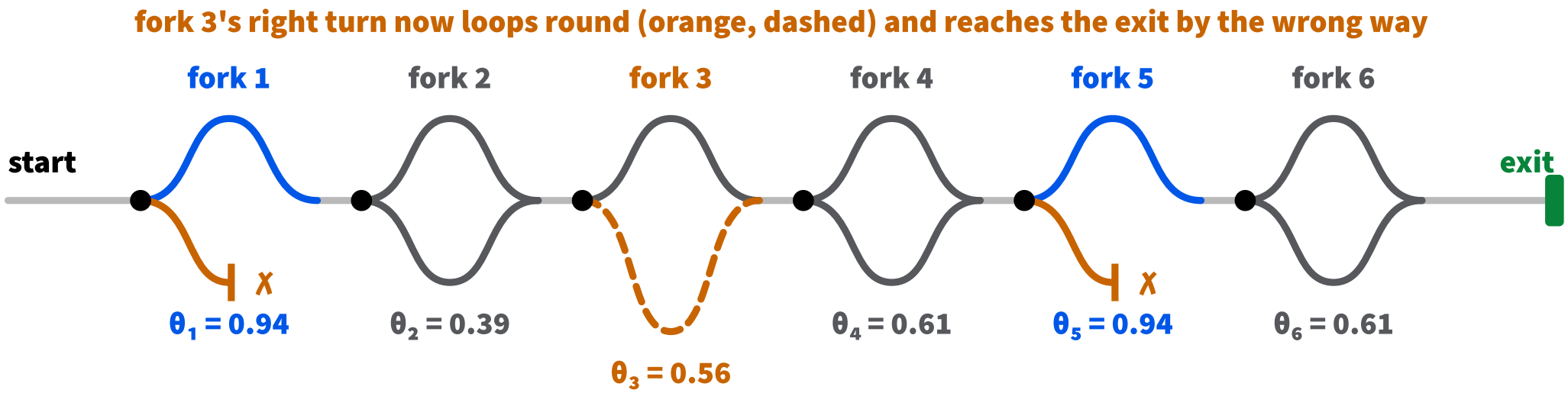
16 walks a round, 10 rounds: kept 0 0 0 0 0 0 0 0 0 0 of 16: nothing to count, the loop stalls

4 096 walks a round: kept 1, 31, 2823 in rounds 1, 2, 3: one lucky walk, then the loop moves

Read: at habits ½ one walk in $2^{12} = 4\,096$ exits, so 16 walks keep nothing and the habits cannot move **(Lecture 6, p. 57)**. The cure: easier problems first.

4 · Where it fails: a shortcut

Trouble: **wrong reasoning, right answer, kept and learned.**



round 10: 16 of 16 walks kept, 7 of them through the loop; after 10 rounds $\theta_3 = 0.56$, not near 1

Read: the answer check keeps every walk that exits, so walks through the loop are learned too.

Q: How would you catch it? A: Check the steps too: keep a walk only if it turned left at fork 3; then $\theta_3 = 0.94$. This is process supervision [Lightman et al., 2023].

5 · The same loop in today's models

Result: the same loop trains today's reasoning models.

system	walk	keep	count
STaR: Self-Taught Reasoner	write one chain of thought per question	the chains whose answer is right	fine-tune on them; repeat
ReST-EM: Reinforced Self-Training	sample many chains per question	the chains that pass a check	fine-tune the base model; a few rounds
Llama 3	sample several answers per prompt	the best, by a learned judge	fine-tune; next round
DeepSeek-R1	sample chains from a trained model	the correct, readable ones	fine-tune, then more reinforcement learning

Read: the maze's two lessons carry over: easy questions first against the cold start, a check of the steps against the shortcut.

This is STaR [Zelikman et al., 2022] and ReST-EM [Singh et al., 2024]; Llama 3 [Grattafiori et al., 2024] and DeepSeek-R1 [DeepSeek-AI, 2025] run a rejection-sampling stage.

Conclusion

TROUBLE The steps are **hidden**, and there are too many paths for the guess step of expectation–maximization (EM).

FIX Way 2: **walk, keep, count**. The kept walks follow p(path | exit), so counting them is the count step of EM.

RESULT The steps that decide the answer are learned; a cold start stalls; a shortcut is learned too, so **check the steps**.

Remember: a model learns only what the answer check can see.

Your turn

Predict, then run the maze.

Experiment 1: the maze walker (`python lab6_explore.py 1`, course site → Week 6 → Lab files)

- 64 walks a round instead of 16: do the free forks wander less?
- Fork 1 starts at habit 0.1: how many rounds until it is found?

Other problems, experiments 2 to 6

2. A noisy picture: Ways 2, 3 and 4.
3. The bits in your phone: messages.
4. Reading a library: guess alone.
5. Ranking players: one match at a time.
6. Messages that swing: damping.

Tip: write your guess first; surprises teach.

Selected References

DeepSeek-AI (2025).
DeepSeek-R1: incentivizing reasoning capability in large language models via reinforcement learning.
arXiv:2501.12948.

Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977).
Maximum likelihood from incomplete data via the EM algorithm.
Journal of the Royal Statistical Society, Series B (Methodological), 39(1):1–38.

Grattafiori, A., Dubey, A., Jauhri, A., et al. (2024).
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arXiv:2407.21783.

Lightman, H., Kosaraju, V., Burda, Y., et al. (2023).
Let's verify step by step.
arXiv:2305.20050.

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Beyond human data: scaling self-training for problem-solving with language models.
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A Monte Carlo implementation of the EM algorithm and the poor man's data augmentation algorithms.
Journal of the American Statistical Association, 85(411):699–704.

Zelikman, E., Wu, Y., Mu, J., and Goodman, N. D. (2022).
STaR: bootstrapping reasoning with reasoning.
In Advances in Neural Information Processing Systems 35.

Thank you!

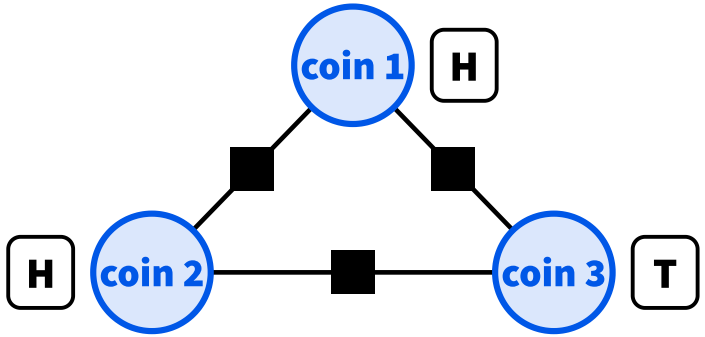
Next: Lecture 7. Way 2 returns whenever a model can only be run forward

Backup: the ring, which way can run, the count by hand, the imaginary walks, exact against sampled, other problems.

Backup: Lecture 6's ring

Eight stories were listed; Way 2 is the one that needs no list.

hidden: three coins, A or B
seen: the flips H, H, T
links: keep ¾ on every link



the question: coin 1's chance of being A



Look: Way 2 replayed the game, kept the runs that showed H, H, T and counted **(Lecture 6, p. 29)**; it is noisy but right on average.

Backup: which way can run?

Only Way 2 runs when the only tool is the model.

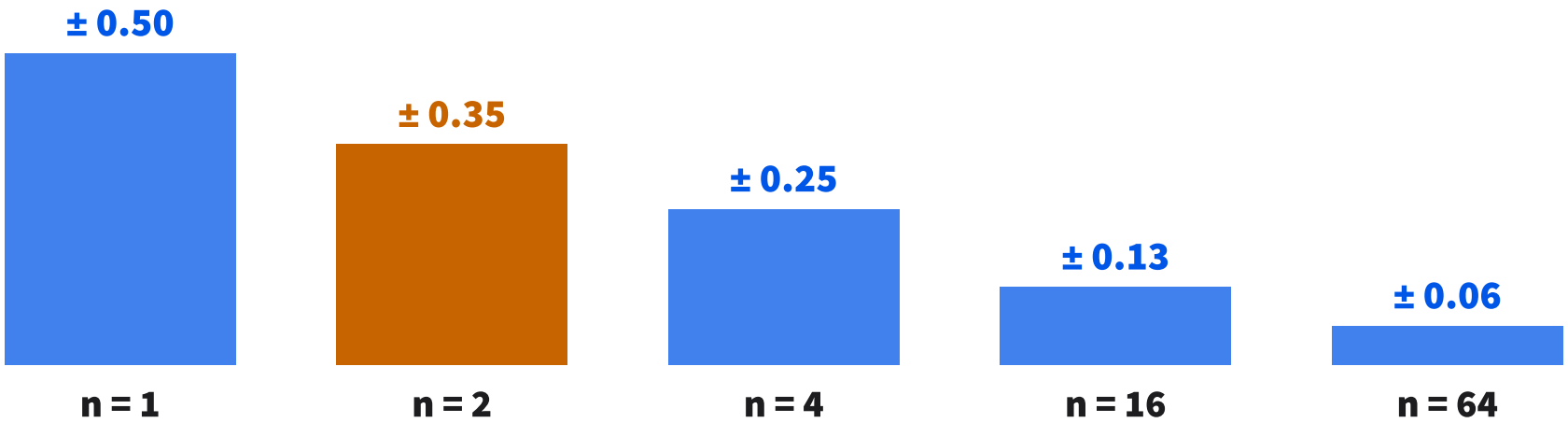
way	what it needs	in a reasoning model?
1 · messages along a line	a line: each step touches only the step before	**no: there is no line**
2 · act it out	running the model forward, and the answer check	**yes: run it forward**
3 · everyone guesses alone	each step touching only a few others	**no: each touches all**
4 · messages in a circle	small links between a few steps	**no: each touches all**

Read: a language model writes the next step from everything written so far, so every step touches all earlier ones: no line for Way 1 **(Lecture 6, p. 27)**, no small links for Ways 3 and 4.

Backup: why the imaginary walks

Two kept walks make a noisy count: add one imaginary walk each way.

typical error of a counted chance (truth ½) from n kept walks: √(½ · ½ / n)



round 1, fork 6: both kept walks turned left: raw count 2 / 2 = 1.00, as if fork 6 were sure

with one imaginary walk each way: $\theta_6 = (2 + 1) / (2 + 2) = 0.75$, held back from certainty

Read: by Theorem 7 **(Lecture 6, p. 31)** the error falls like $1/\sqrt{n}$, so two walks say little; and a habit of exactly 1 could never change again.

Backup: exact EM against sampling

Exact expectation–maximization (EM) over all 64 paths agrees where it matters.

	exact EM, all 64 paths		walk, keep, count	
fork	**round 1**	**round 10**	**round 1**	**round 10**
1, 3, 5 (matter)	0.75	**0.93**	0.75	**0.94**
2 (free)	0.50	0.50	0.50	0.67
4 (free)	0.50	0.50	0.75	0.61
6 (free)	0.50	0.50	0.75	0.50

Read: exact EM uses the same count, with the kept walks replaced by their average over all 64 paths: the forks that matter agree, and the free forks stay at ½ while sampling wanders, the noise of Theorem 7 **(Lecture 6, p. 31)**.

Backup: the count step of round 1 by hand

Round 1: two kept walks, L R L L L L and L L L L L L.

$$\theta_t \leftarrow \frac{\text{left turns at fork } t \text{ among the kept walks} + 1}{\text{kept walks} + 2}$$

> the count step (Lecture 6, p. 30)

$$\theta_1 = \frac{2+1}{2+2} = 0.75, \qquad \theta_2 = \frac{1+1}{2+2} = 0.50$$

> fork 1: both turned left; fork 2: one did

$$\theta_3 = \theta_4 = \theta_5 = \theta_6 = \frac{2+1}{2+2} = 0.75$$

> both turned left at forks 3 to 6

$$\text{without the imaginary walks: } \theta_1 = \frac{2}{2} = 1.00$$

> a habit at 1 could never move again

Notice: the exact guess step gives 1 at forks 1, 3, 5 and ½ at 2, 4, 6; two walks cannot tell fork 3 from fork 4.

Backup: Theorem 7's error for n kept walks

A counted chance p from n kept walks is off by about √(p(1 − p)/n).

kept walks n	p = ½: typical error	p = 0.9: typical error	where it happens in the deck
1	± 0.50	± 0.30	the cold start with 4 096 walks, round 1
2	± 0.35	± 0.21	round 1 (p. 9)
4	± 0.25	± 0.15	the shortcut maze, round 1 (p. 12)
16	± 0.13	± 0.08	rounds 7 and 10 (p. 10)
64	± 0.06	± 0.04	the kit, 64 walks a round

Notice: four times the kept walks halve the error (Lecture 6, p. 31); the free forks wander by about this much every round, which is the wandering seen over ten rounds.

Backup: five other troubles, in the kit

Five other troubles, in the kit as experiments 2 to 6.

	Trouble	Fix	Where
2	a noisy picture	redraw, guess alone or messages (Ways 2–4)	photos, scans
3	the bits in your phone	messages in a circle (Way 4)	phones, Wi-Fi, space
4	reading a million documents	guess alone, then count (Way 3, then count)	search
5	ranking players from wins	the truth judges one match at a time	matchmaking
6	messages that swing	damping: half old, half new	any cycles

Read: each is one of Lecture 6's four ways on a deployed system; the kit's README states a prediction for each, and the answers after it.